

Project Design Phase-II

Data Flow Diagram (Exquisite Granularity)

Date	03 June 2026
PRN	2023011032040
Project Name	AI-Based Threat Intelligence Platform
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

AI-Based Threat Intelligence Platform - Level 1 DFD Blueprint

This diagram separates the logical flow of data from the physical architecture. Instead of looking at servers and modules, this traces the exact lifecycle and mutation of a data packet from ingestion to threat mitigation.

1. External Entities (Sources/Destinations)

- E1: ABC Corp IT Infrastructure (Raw generators: SIEM, Endpoints, Network Switches)
- E2: Global Threat Intelligence Feeds (STIX/TAXII, OSINT, Dark Web Scrapers)
- E3: SOC (Security Operations Center) Analyst
- E4: Automated Defense Systems (SOAR, Firewall APIs, Active Directory)

2. Processes (Data Transformation)

- 1.0: Ingest & Parse Telemetry (Converts raw proprietary logs into standardized JSON)
- 2.0: Contextual Data Enrichment (Matches IPs/Hashes with known bad actors and vulnerabilities)
- 3.0: Execute ML Anomaly Detection (Runs behavioral models against enriched data vectors)
- 4.0: Calculate Risk & Attribute Threat (Scores the anomaly and links to specific TTPs)
- 5.0: Generate Mitigation Playbooks (Creates actionable UI dashboards and automated response scripts)

3. Data Stores

- D1: Raw Telemetry Lake (Unprocessed PCAP & Syslogs)
- D2: Threat Intelligence Graph DB (Relational database of known malware and threat actors)

- D3: ML Baseline Models (Stored profiles of 'normal' employee/network behavior)
- D4: Active Incident Ledger (Database of currently tracked threats and their statuses)

4. Exquisite Data Flows (The Arrows)

--- Ingestion Phase ---

- [Raw PCAP & Syslogs] -> From [E1: ABC Corp IT] to [1.0: Ingest & Parse]
- [STIX/TAXII Threat Indicators] -> From [E2: Global Threat Feeds] to [2.0: Contextual Enrichment]
- [Standardized JSON Logs] -> From [1.0: Ingest & Parse] to [D1: Raw Telemetry Lake]

